Lightship

SOW#3
AI MVP

Kickoff

05/03/26

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AGENDA

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Background

Lightship is a driver training and safety technology company. Lightship develops interactive training content from dashcam footage to improve driver safety and evaluation. In the PoC phase, we validated an automated pipeline for hazard detection and frame annotation. The MVP phase enhances detection accuracy with pre-trained models, adds LLM-powered video classification, and expands coverage to construction sites.

MVP Goal

AI-Powered Video Annotation & Classification System

Primary Goal / Objective:
Enhance Lightship's automated dashcam annotation pipeline with superior pre-trained detection models and LLM-powered video classification, significantly improving labeling accuracy and enabling intelligent content categorization.

Expected Outcomes:
Significantly improved detection accuracy for motorcycles, lanes, road signs, and construction objects.
Automated video classification into 4 training types using LLM analysis.
Auto-generated JSON configs tailored to each video type (reactivity, Q&A, hazard, construction).

Target Metrics:
≥80% Motorcycle Detection Recall on validation set
≥80% Lane Detection Accuracy (IoU score)
≥80% Road Sign Detection Precision
≥80% Construction Object Precision
≥80% Video Classification Accuracy (4 classes)

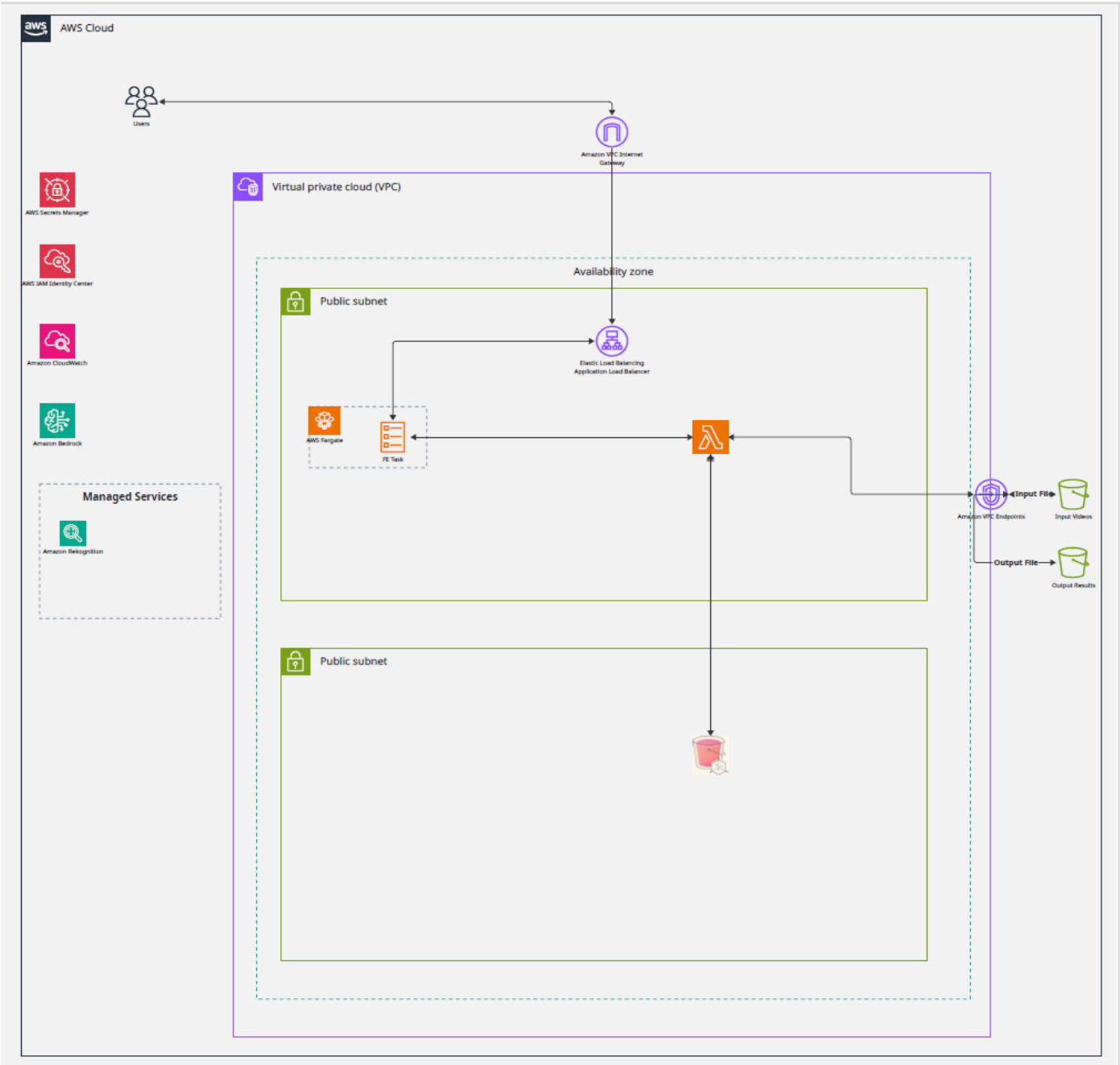
Activities & Deliverables

- **Project kick-off and planning**
  - Initial kick-off meeting
  - Review Project requirements and timelines
  - Define collaboration tools and/or processes
- **Design**
  - Jointly define MVP evaluation criteria and success metrics
  - Select and benchmark pre-trained models (Detectron2, Rekognition, YOLOv12 XL)
    - Develop MVP
    - Integrate pre-trained models for motorcycle, lane, road sign, and construction detection
    - Implement LLM-powered video classification (4 types) and JSON config generation
    - Add batch processing, custom S3 output paths, and single image mode
    - Implement cross-frame object tracking and hazard assessment
    - Optimize detection for rainy/adverse weather conditions
- **Deliver**
  - Provisioned MVP solution with enhanced detection pipeline
  - Comprehensive documentation and handover session
  - Performance evaluation report (PoC vs MVP comparison)

MVP SCOPE

Feature/Screen	Description
Pre-trained Model Research	Evaluate and select best models per category (Detectron2, Rekognition, YOLOv12 XL)
Enhanced Detection Pipeline	Motorcycle, lane (Detectron2), road sign (50+ types), and weather-aware detection
Construction Detection	Pre-trained construction object detection with 15+ labels and safety prioritization
LLM Video Classification	Classify videos into 4 types and auto-generate class-specific JSON configs
Cross-Frame Tracking	Object tracking across frames with trajectory analysis and hazard assessment
Batch Processing	SQS-based async processing for multiple video uploads with completion notifications
Custom S3 Output Path	User-defined output paths for organized result storage
Single Image Mode	Upload single frame for construction site object detection
Evaluation & Testing	Performance benchmarks comparing PoC vs MVP across all detection categories
Documentation & Handover	Technical docs, architecture diagrams, optimization guides, and debugging guides

MVP Architecture



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Data Requirements

Data Sources:

- 30-50 sample dashcam videos (motorcycle, lanes, weather, construction scenarios)
- 30-50 labeled frames as ground truth for validation
- Video class definitions and taxonomy (reactivity, Q&A, hazard, construction)
- Educational content templates for Q&A generation
- Construction site object taxonomy

Data Availability & Access:

- Client to provide sample videos within 1 week of kickoff
- Client to provide 10 labeled frames per video for validation
- AWS account access with Bedrock, Rekognition, and S3 permissions required

Team & Timeline

Commit MVP Delivery Team:

- Daniel Nahman - Product Manager - IST
- Ido Garbi - Head of AI - IST
- AI Developer – Sagi Levi + Asaf Lavi - IST
- Adi Ben David- DevOps – IST
- Niv Spector- Project Manager -IST
- Alex Coggins - Client Partner - UKT

Lightship Stakeholders:

- Zechari Tempesta- CEO

Sprint Start Date	5 Mar 2026
Sprint End Date	19 Mar 2026
Tentative MVP Presentation Date	30 Apr 2026

We estimate the development process will last 6 weeks (3 Sprints).

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SOW Details

- SoW2 signed Mar 2026
- $55K fixed cost MVP
  - AWS funding TBD
- Client facing kickoff scheduled: 5 Mar 2026
- Aim to complete this project in no more than 6 weeks from start date

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Next Steps

- Client facing kickoff
  - What do you need from this?
    - AWS account access (Bedrock, Rekognition, S3)?
    - GitHub access?
    - Sample dashcam videos (30-50)?
    - Labeled ground truth frames?
    - Video class taxonomy and definitions?
- Review resources provided ahead of development start

Thank You

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